

JOÃO YAKUBETS

Game Developer — Unity / C#

Braga, Portugal · +351 934 588 164 · joao.yakubets@hotmail.com

github.com/unduke2 · hypnagogic.itch.io/ascensionexe

PROFILE

Unity developer with 2+ years of hands-on experience — gameplay mechanics, systems, UI, and shaders — with particular depth in systems architecture and performance (DOTS/ECS, Burst, dependency injection). Currently in production on a retro FPS targeting commercial release, while deliberately deepening technical art (shader writing, 3D modelling) through a dedicated capstone. Game-jam winner; founder of a commercially shipped product.

EXPERIENCE

Game Developer & Co-founder — Hypnagogic Arts

2025 – Present · Braga

4-person indie team (programming, composer, 2D artist, writer) · Unity / C#

- Developing **Valley of Screams**, a retro FPS in Unity — gameplay mechanics, enemy AI, and systems architecture, in production toward a commercial release ahead of an EU funding application (Feb 2027).
- Designed and built a composable agent framework: coded behaviours configured as ScriptableObject assets and assembled drag-and-drop without code — powering weapons, enemies, encounters, and player systems; architected on dependency injection (Zenject).
- Won PSX Horror Jam (2026) with **Ascension.exe**, a short atmospheric horror experience — sole programmer in a team of 4, building all gameplay and systems around the team's art, story, and music. Published on itch.io.
- Lead all programming and technical development, collaborating with a composer, 2D artist, and writer from concept through publishing.

Founder — NextStint (nextstint.app)

Apr 2026 – Present

- Designed, shipped, and operate a commercial offline-first PWA for EU transport compliance (Stripe billing, 3 languages, live in production) — built AI-first with Claude Code: sole owner of product design, architecture decisions, specification, verification, and support.

SELECTED PROJECT

Space-Drone-Simulator — Unity DOTS/ECS Performance Benchmark

github.com/unduke2

- 20,000-drone steering swarm implemented twice — ECS/Burst/Jobs vs. idiomatic MonoBehaviours — sharing identical logic and spatial-hash neighbour queries, isolating architecture as the only variable; up to 21.7× frame-time advantage (13.2× like-for-like).
- Validated through a reproducible benchmark methodology (fixed camera poses, warm-up windows, median/p95/p99 sampling) across 23 instrumented builds — full write-up on GitHub.

SKILLS

- **Engine & languages:** Unity (URP, DOTS/ECS, Burst, Jobs, Entities Graphics), C#
- **Full-cycle development:** gameplay mechanics, enemy AI, UI, systems integration, content, builds & publishing
- **Architecture:** dependency injection (Zenject), ScriptableObject-driven data design, composable/modular systems
- **Graphics & VFX:** Shader Graph, VFX Graph, Particle System, HLSL (learning), Blender — capstone in production: ice-cave environment with custom shaders (transmission, mesh blending)
- **Performance:** profiling (Unity Profiler), benchmarking methodology, data-oriented design
- **Tools:** Git/GitHub · Blender · ComfyUI (Stable Diffusion pipelines) · Claude Code (AI-assisted development: specification, verification-driven review, test harnesses)
- **Languages:** Portuguese (fluent), English (fluent)

EDUCATION

Restart — Criação e Produção de Videojogos

Oct 2023 – 2024

One-year intensive programme in game creation and production

Universidade do Minho — Engenharia e Gestão de Sistemas de Informação

2020 – 2023

Coursework completed toward degree (programme not concluded)